

Bayesian Logistic Regression Model (BLRM) in SAS

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Brief introduction to BLRM

Base model (Single dose)

In single dose model, we assume that for dose level d , the number of patients with DLT rates (r_d) in a cohort of size n_d is binomial, that is

$$r_d | n_d \sim \text{Bin}(\pi_d, n_d),$$

with DLT rates

$$\text{logit}(\pi_d) = \log(\alpha) + \beta \log\left(\frac{d}{d^*}\right),$$

where d^* is an arbitrary selected scaling dose level, called **reference dose**.

A brief interpretation of the parameters α and β :

- α : the odds, $\frac{\pi}{1-\pi}$, at reference dose level.
- β : the increase in the log of odds of a DLT rate by a unit increase in log of dose.

Generally, we use a bivariate normal distribution as the prior distribution for the model parameters.

$$(\log \alpha, \log \beta) \sim \text{N}(m, S),$$

where m is the mean vector of $(\log \alpha, \log \beta)$ and S is the covariance matrix.

Combination model

Let $\pi_1(d_1)$ be the probability of DLT at dose d_1 for drug A, and $\pi_2(d_2)$ be the probability of DLT at dose d_2 for drug B. The model is for each drug is

$$\text{logit}(\pi_1(d_1)) = \log(\alpha_1) + \beta_1 \log\left(\frac{d_1}{d_1^*}\right)$$

$$\text{logit}(\pi_2(d_2)) = \log(\alpha_2) + \beta_2 \log\left(\frac{d_2}{d_2^*}\right)$$

where d_1^* and d_2^* are the reference dose for the drug which is used to normalize the dose d_1 and d_2 respectively, and the prior for $\theta_1 = (\log \alpha_1, \log \beta_1)$ is $\text{N}(m_1, S_1)$ and the prior for $\theta_2 = (\log \alpha_2, \log \beta_2)$ is $\text{N}(m_2, S_2)$.

The toxicity $\pi(d_1, d_2)$ for the combination is modeled as the “no-interaction” DLT rate

$$\tilde{\pi}(d_1, d_2) = 1 - (1 - \pi_1(d_1))(1 - \pi_2(d_2)) = \pi_1(d_1) + \pi_2(d_2) - \pi_1(d_1)\pi_2(d_2)$$

Written in Odds form:

$$\text{Odds}(\tilde{\pi}(d_1, d_2)) = \text{Odds}(\pi_1(d_1)) + \text{Odds}(\pi_2(d_2)) + \text{Odds}(\pi_1(d_1)) \times \text{Odds}(\pi_2(d_2))$$

Since $\text{Odds}(\pi) = \frac{\pi}{1-\pi}$.

Adding the interaction term:

$$\text{Odds}(\pi(d_1, d_2)) = \text{Odds}(\tilde{\pi}(d_1, d_2)) \times g(\eta, d_1, d_2)$$

g is called odd-multiplier, which should fulfill the constraints: $g(\eta, d_1, 0) = g(\eta, 0, d_2) = 1$.

Different odd-multiplier:

1. Linear

$$g_{\text{linear}}(\eta, d_1, d_2) = \exp\left(\eta \frac{d_1}{d_1^*} \frac{d_2}{d_2^*}\right)$$

$$\text{logit}(\pi(d_1, d_2)) = \text{logit}(\tilde{\pi}(d_1, d_2)) + \eta \frac{d_1}{d_1^*} \frac{d_2}{d_2^*}$$

2. Saturating

$$g_{saturating}(\eta, d_1, d_2) = \exp\left(2\eta \frac{\frac{d_1}{d_1^*} \frac{d_2}{d_2^*}}{1 + \frac{d_1}{d_1^*} \frac{d_2}{d_2^*}}\right)$$

$$\text{logit}(\pi(d_1, d_2)) = \text{logit}(\tilde{\pi}(d_1, d_2)) + 2\eta \frac{\frac{d_1}{d_1^*} \frac{d_2}{d_2^*}}{1 + \frac{d_1}{d_1^*} \frac{d_2}{d_2^*}}$$

For η , we assume that $\eta \sim N(m_\eta, S_\eta)$.

For more information about the interaction term, one can refer to Widmer et al. (2023).

Workflow (Single Dose version)

Before starting the trial, one needs to at least decide:

- A list of dose levels in same frequency, such as (2mg, 5mg, 10mg, 20mg, 25mg) QD,
- A reference dose level selected from the list above, like 10mg QD,
- The parameter of prior distribution,
- The starting dose level (may not necessary be the lowest dose level),
- The under-dosing interval, such as $[0, 0.16)$,
- The target toxicity interval, such as $[0.16, 0.33)$,
- The over-dosing interval, such as $[0.33, 1]$,
- The boundary b for the principle of **escalation with overdose control** (EWOC): For certain dose level $\mathbf{d}$, the posterior probability of over dosing π_d should not be too high, that is, $\Pr(\pi_d \in \text{over-dosing interval} | \text{Data}) \leq b$,
- Maximum sample size,
- The cohort size for each enrollment,
- Whether allow for dose skipping in escalation, that is, when current dose level is 2mg QD, can 10mg or higher dose level be selected as the next dose level?
- Stopping rule:
 - Reached maximum sample size
 - All dose levels do not satisfy EWOC, i.e. $\forall d \in \text{dose list}, \Pr(\pi_d \in \text{over-dosing interval} | \text{Data}) > b$
 - Some other custom rules

When the trial starts, a cohort of subjects in starting dose will be enrolled, each time a cohort is enrolled and the DLT is observed, one can update the blrm model and output the posterior probability of toxicity in each dose level, and then choose the dose level d_i such that satisfy the EWOC principle and with the highest posterior probability of toxicity in the target toxicity interval following the dose skipping rule, that is, select d_i such that

$$\arg \max_i \Pr(\pi_{d_i} \in \text{target toxicity interval} | \text{Data})$$

and $\Pr(\pi_{d_i} \in \text{over-dosing interval} | \text{Data}) \leq b$. Repeat this process until the pre-specified rule is met.

Selection of parameters

Prior distributions As a bayesian model, BLRM is sensitive to the selection of prior distribution, to describe the detail of selecting prior distributions is out of the scope of this report, therefore only brief introduction is applied here.

In Neuenschwander, Branson, and Gsponer (2008), the authors suggested to derive a weakly-informative BVN prior based on quantiles from minimally informative unimodal Beta distributions when no relevant historical DLT data available, also, in Zhao and Yang (n.d.), a simpler way to derive a weakly-informative BVN prior is provided.

When history information or concurrent trial data are available, Neuenschwander, Roychoudhury, and Schmidli (2016) suggested the Meta-Analytic-Predictive (MAP) Priors.

The boundary for EWOC The selection of boundary b for EWOC will also affect the performance of BLRM, briefly speaking, a low value for the boundary makes the model too conservative which leads to a selection of dose level that is much lower than MTD, however, a higher value for the boundary makes the model too aggressive, even though the accuracy will be higher, but more enrolled subjects will be exposed to over-toxic dose level in a trial, in Zhou, Yuan, and Nie (2018), the performances of BLRM with and without EWOC is compared. Besides, in Guo et al. (2025); Zhang, Chiang, and Wang (2022), some methods to determine an optimal boundary are suggested.

SAS Macros for Bayesian Logistic Regression Model (BLRM) Dose-Finding

Part 1: Conducting a Phase I Dose-Finding Trial

Overview The first part enables running a single dose-finding trial using BLRM methodology. Key components:

1. **Model Fitting:** Bayesian logistic regression with MCMC
2. **Dose Selection:** EWOC-based decision rules
3. **Output:** Posterior summaries and dose recommendations

blrm_main Fits the BLRM model using PROC MCMC in SAS.

Macro introduction:

```
%blrm_main(
  datain    = ,           /* Current trial data */
  dataout   = ,           /* Output data */
  ref_dose  = ,           /* Reference dose, for scaling the dose range*/
  mu1       = ,           /* Prior mean log(alpha) */
  mu2       = ,           /* Prior mean log(beta) */
  v1        = ,           /* Prior variance log(alpha) */
  v2        = ,           /* Prior variance log(beta) */
  rho       = ,           /* Prior covariance */
  alpha     = 0.05,       /* the alpha level for the equal-tail and HPD intervals*/
  nbi       = 2000,       /* Burn-in iterations */
  nmc       = 5000,       /* Sampling iterations */

  simming   = 0,          /* Simulation indicator, if set to 1, the ods will close */
  seed      = 9527        /* Seed */
);
```

Note: for the input dataset **datain** in **blrm_main**, we require the variable names as follow:

- **dose** : the dose level, in numeric form
- **n** : the enrolled subject number of the dose level
- **dltn**: the number of dlt subjects of the dose level

When **debug** is set to 0 and **simming** is set to 1, **blrm_main** outputs 1 main dataset, **dataout=out_mcmc**, which contains the sampling from posterior distribution of $\log(\alpha)$ and $\log(\beta)$.

Obs	dose	n	dltn
1	1	3	0
2	5	6	1
3	10	5	2

blrm_stat Output brief statics from **blrm_main**. **Macro introduction:**

```
%blrm_stat(
  dose_list = ,          /* the dose levels wanted to summarize the posterior dlt, */
                        /* not necessary be the same dose levels in blrm_main. */
  dataout   = ,          /* Output data */
  ref_dose  = ,          /* Reference dose, for scaling the dose range*/
  mcmcout   = ,          /* The output dataset from blrm_main */
  stat      = ,          /* The statistics wanted to summarize, */
                        /* compatible to proc means only */
  debug     = ,          /* Debug flag, if set to 1, median dataset will be deleted */
);
```

blrm_ewoc Summarize the ewoc information from blrm_main.

Macro introduction:

```
%blrm_ewoc(
  dose_list = ,          /* the dose levels wanted to summarize the posterior dlt, */
                        /* not necessary be the same dose levels in blrm_main. */
  dataout   = ,          /* Output data */
  ref_dose  = ,          /* Reference dose, for scaling the dose range*/
  mcmcout   = ,          /* The output dataset from blrm_main */
  ud        = ,          /* The under dosing boundary */
  od        = ,          /* The over dosing boundary */
  ewoc      = ,          /* The escalation with overdose control RULE */
  debug     = ,          /* Debug flag, if set to 1, median dataset will be deleted */
);
```

Examples Firstly, assume that we have a dose escalation trial, with dose levels: 1mg, 5mg, 10mg, 12mg and 15mg. Using the 10mg as reference dose, and then we already have 3 subjects enrolled in dose level 1mg, no dlt is observed, and 6 subjects enrolled in dose level 5mg, 1 dlt is observed, and the current dose level is 5mg QD.

```
option locale='en_us';
data dlt_dummy;
  dose=1;n=3;dlt=0;output;
  dose=5;n=6;dlt=1;output;
run;
```

```
proc print data=dlt_dummy noobs;run;
```

dose	n	dlt
1	3	0
5	6	1

Using **blrm_main** we can update the Logistic model, when parameter **simming** is set to 0, the ods will output the PROC MCMC result.

```
%blrm_main(datain=dlt_dummy,
  dataout=out_mcmc,
  ref_dose=10,
  /*mu1=log(0.2/(1-0.2)), */
  mu1 = -4,
  mu2=0.5,
  v1=4,
```

```
v2=1, rho=0,alpha=0.05, nbi=2000, nmc = 10000, seed = 9527,SIMMING=1);
ods exclude close;
```

When debug is set to 0 and simming is set to 1, blrm_main outputs 1 main dataset, dataout=out_mcmc, which contains the sampling from posterior distribution of $\log(\alpha)$ and $\log(\beta)$.

```
proc print data=out_mcmc(obs=5) noobs;run;
```

Iteration	log_ alpha	log_beta	LogPrior	LogLike	LogPost
2001	-1.2041	-0.8748	-4.4533	-1.2436	-5.6968
2002	-1.2041	-0.8748	-4.4533	-1.2436	-5.6968
2003	-1.2041	-0.8748	-4.4533	-1.2436	-5.6968
2004	-0.0455	0.5853	-4.4895	-1.0027	-5.4922
2005	-2.1173	-0.7835	-3.7978	-1.2701	-5.0679

Using this output, one can summarize some simple statistics of predicted toxicity of each dose level using blrm_stat macro.

```
%blrm_stat(dose_list=%str(1,5,10,12,15),ref_dose=10,mcmcout=out_mcmc,
dataout=d1,debug=0,stat=%str(mean, std, q1, median, q3));
proc print data=d1 noobs;run;
```

dose	Mean	StdDev	Q1	Median	Q3
1	0.02279	0.03715	0.001703	0.00897	0.02696
5	0.07861	0.08072	0.020736	0.05137	0.10870
10	0.16618	0.15859	0.045360	0.11346	0.24024
12	0.19996	0.18874	0.053811	0.13707	0.28283
15	0.24460	0.22469	0.064733	0.17044	0.36352

To decide the next dose level to enroll subject, user can use the blrm_ewoc macro to summarize the predicted toxicity of each dose level in EWOC sense, in this example, we set the under dosing interval as [0.0,0.16), the target toxicity interval as [0.16,0.33) , and the over dosing interval as [0.33,1], also, the boundary of EWOC is set to 0.25.

```
%blrm_ewoc(dose_list=%str(1,5,10,12,15),ref_dose=10,mcmcout=out_mcmc,dataout=ewoc_next,
ud=0.16,od=0.33,ewoc=0.25);
proc print data=ewoc_next noobs;run;
```

dose	UD_Mean	TT_Mean	OD_Mean	EWOC
1	0.9868	0.0123	0.0009	True
5	0.8644	0.1192	0.0164	True
10	0.6171	0.2358	0.1471	True
12	0.5479	0.2430	0.2091	True
15	0.4816	0.2409	0.2775	False

We can see that both 10mg and 12mg satisfy the EWOC principle, and dose level 12mg has the highest predicted probability in target toxicity interval, however, if the trial does not allow dose skipping, one can only select 10mg as the next dose level to enroll subjects.

Suppose the 12mg is chosen as the next dose level, for some practical reason 4 subjects are enrolled and 3 of them had experienced DLT, then we can update the model as follow

```
data dlt_new;
dose=12;n=4;dlt=3;
```

```

run;
data dlt_dummy_new;
  set dlt_dummy
      dlt_new;
run;

%blrm_main(datain=dlt_dummy_new,
           dataout=out_mcmc,
           ref_dose=10,
           /*mu1=log(0.2/(1-0.2)), */
           mu1 = -4,
           mu2=0.5,
           v1=4,
           v2=1, rho=0,alpha=0.05, nbi=2000, nmc = 10000, seed = 9527,SIMMING=1);

```

Part 2: Simulation

Overview The second part evaluates trial design operating characteristics through simulation:

1. **Trial Replication:** Run hundreds/thousands of virtual trials
2. **Performance Metrics:** Calculate dose selection accuracy
3. **Safety Assessment:** Measure patient exposure to toxic doses

blrm_sim_one Runs a single simulated trial, the simulation will stop when one of the condition below is met:

1. All sample size are used up,
2. All dose levels not satisfied EWOC.

Macro introduction:

```

%blrm_sim_one(
  dose_list  = ,
  ref_dose   = ,
  mu1        = ,
  mu2        = ,
  v1         = ,
  v2         = ,
  rho        = ,
  alpha      = 0.05,
  nbi        = ,
  nmc        = ,
  ud         = ,
  od         = ,
  ewoc       = ,
  true_tox   = true_tox,    /* Dataset with true probs */
  start_dose = ,            /* Starting dose */
  max_sample = ,            /* Max patients */
  cohortn    = ,            /* Cohort size */
  skip       = ,            /* Allow dose skipping? */
  sim_batch  = ,            /* Simulation ID */
  seed       = 12345,       /* Random seed */
  debug      = 0
);

```

A example of true toxicity dataset, the same variable names are required:

dose	true_tox
10	0.39
20	0.45
40	0.48
50	0.49

When `debug` is set to 0, the `blrm_sim_one` will output 3 dataset:

1. `DLT_RECORD`: A dataset that record the dlt count in each time of enrollment, use `enroll_batch` as ID,
2. `EWOC_RECORD`: A dataset that record the EWOC situation after each time of enrollment, use `enroll_batch` as ID,
3. `SIM_RECORD`: A dataset that record the selected dose level and the used sample in this simulation.

`blrm_sim_n` Runs multiple simulations, using `sim_batch` as ID to identify each single simulation.

Macro introduction:

```
%blrm_sim_n(
    dose_list = ,
    ref_dose  = ,
    mu1       = ,
    mu2       = ,
    v1        = ,
    v2        = ,
    rho       = ,
    alpha     = 0.05,
    nbi       = ,
    nmc       = ,
    ud        = ,
    od        = ,
    ewoc      = ,
    true_tox  = true_tox,
    start_dose = ,
    max_sample = ,
    cohortn   = ,
    skip      = ,
    sim_time  = 5000,          /* Number of simulations */
    seed      = 12345,
    time_out  = time_summary, /* Output for used time data */
    time_sum  = 1              /* If set to 1, ods will output the time summary */
);
```

When `debug` is set to 0, the `blrm_sim_n` will output 3 dataset:

1. `DLT_ALL`: A dataset that record the dlt count in each time of enrollment in each simulation batch,
2. `EWOC_ALL`: A dataset that record the EWOC situation after each time of enrollment in each simulation batch,
3. `SIM_ALL`: A dataset that record the selected dose level and the used sample in each simulation batch,
4. `TIME_RECORD`: A dataset that record the starting time, end time and used time of each simulation batch in second,
5. `TEMP_TIME`: A dataset that record the starting time, end time and used time of the whole simulation in second,
6. `TIME_SUMMARY`: A data set that set `TIME_RECORD` and `TEMP_TIME` together.

`blrm_sim_sum` Summarizes simulation results, and output:

Macro introduction:

```
%blrm_sim_sum(  
    sim_data=,          /* Output from blrm_sim_n (sim_all) */  
    dlt_data=,          /* Patient-level data from blrm_sim_n (dlt_all) */  
    true_tox=,          /* Dataset with true toxicity probabilities */  
    target_range_low=0.16, /* Lower bound of target toxicity range */  
    target_range_high=0.33, /* Upper bound of target toxicity range (also overdose threshold) */  
    acceptable_low=0.05,  /* Lower bound for acceptable dose */  
    out_summary= ,       /* Output dataset name */  
    debug=0  
);
```

Output Metrics The performance_stats dataset contains:

Metric	Description
Prop_Correct_Dose	% trials selecting true target dose
Prop_Acceptable_Dose	% selecting acceptable doses
Prop_Over_Toxic_Dose	% selecting toxic doses
Avg_Patients_Over_Toxic	Average patients exposed to toxic doses

Simulation Examples

1. Use up all sample size Using the same dose level list in the previous part, we construct a dummy toxicity dataset.

```
data dummy_tox;  
    dose=1 ;true_tox=0.10;output;  
    dose=5 ;true_tox=0.15;output;  
    dose=10;true_tox=0.24;output;  
    dose=12;true_tox=0.32;output;  
    dose=15;true_tox=0.63;output;  
run;
```

dose	true_tox
1	0.10
5	0.15
10	0.24
12	0.32
15	0.63

In this example, we set the under dosing interval as $[0, 0.16)$, the target toxicity interval as $[0.16, 0.33)$, and the over dosing interval as $[0.33, 1]$, also, the boundary of EWOC is set to 0.3, and note that the starting dose level 10mg, maximum sample size is 12 and the cohort size is 3, skipping dose is not allowed.

```
option locale='en_us';  
%blrm_sim_one(dose_list=%str(1,5,10,12,15),  
    ref_dose=10,mu1=-5, mu2=0, v1=4, v2=1, rho=0 ,  
    alpha=0.05, nbi=1000, nmc = 10000,  
    ud=0.16,od=0.33,ewoc=0.3,  
    true_tox=dummy_tox,  
    start_dose=10,sim_batch=4,  
    max_sample=12,cohortn=3,skip=0,seed=1234,debug=0);  
ods exclude close;
```

```
proc print data = DLT_RECORD noobs;run;
```

dose	n	dltn	true_tox	enroll_ batch	sim_ batch
10	3	0	0.24	1	4
12	3	0	0.32	2	4
15	3	2	0.63	3	4
12	3	2	0.32	4	4

The DLT_RECORD shows that when the trial is stop, the maximum sample size (12) is reached, and after the cohort with 15mg dose level is enrolled, the model decided that the trial should deescalate back to 12mg dose level. The selected dose level and used sample size in this single time simulation is recorded at SIM_RECORD dataset:

```
proc print data = SIM_RECORD noobs;run;
```

dose_ select	sample_ used	sim_ batch
12	12	4

To confirm the reason that the model selected 12mg as MTD, we can check the last enroll batch part in EWOC_RECORD:

```
proc sql noprint;
select max(enroll_batch) into: last_enroll trimmed from EWOC_RECORD;
quit;
proc print data = EWOC_RECORD(where=(enroll_batch=&last_enroll.)) noobs;run;
```

dose	UD_Mean	TT_Mean	OD_Mean	EWOC	enroll_ batch	sim_ batch
1	0.9565	0.0392	0.0043	True	4	4
5	0.8507	0.1316	0.0177	True	4	4
10	0.5540	0.3563	0.0897	True	4	4
12	0.2540	0.5067	0.2393	True	4	4
15	0.0827	0.2940	0.6233	False	4	4

2. Early termination In this part we use the same dose level list, but with an over toxic assumption, this time starts from 5mg dose level,

```
data over_tox;
  dose=1 ;true_tox=0.35;output;
  dose=5 ;true_tox=0.5;output;
  dose=10;true_tox=0.5;output;
  dose=12;true_tox=0.52;output;
  dose=15;true_tox=0.63;output;
run;
```

dose	true_tox
1	0.35
5	0.50
10	0.50
12	0.52
15	0.63

```
option locale='en_us';
%blrm_sim_one(dose_list=%str(1,5,10,12,15),
  ref_dose=1,mu1=-1, mu2=0, v1=4, v2=1, rho=0 ,
  alpha=0.05, nbi=1000, nmc = 10000,
  ud=0.16,od=0.33,ewoc=0.3,
  true_tox=over_tox,
  start_dose=5,sim_batch=4,
  max_sample=12,cohortn=3,skip=0,seed=5744,debug=0);
ods exclude close;
```

DLT_RECORD:

dose	n	dltn	true_tox	enroll_ batch	sim_ batch
5	3	3	0.5	1	4

EWOC_RECORD:

dose	UD_Mean	TT_Mean	OD_Mean	EWOC	enroll_ batch	sim_ batch
1	0.1495	0.1431	0.7074	False	1	4
5	0.0002	0.0049	0.9949	False	1	4
10	0.0001	0.0022	0.9977	False	1	4
12	0.0000	0.0017	0.9983	False	1	4
15	0.0000	0.0011	0.9989	False	1	4

SIM_RECORD:

dose_ select	sample_ used	sim_ batch
.	3	4

In this case, simulated trial is terminated when the DLT is observed in the first batch of enrolled subjects, and selected none of the dose as MTD.

3. Generate n times simulation and summarize The macro `blrm_sim_n` runs `blrm_sim_one` n times with different seed, and stack the results together, therefore the generated output will not be displayed here, instead, the output from `blrm_sim_n` will be summarized using `blrm_sim_sum` in this example, users can analyze the output from `blrm_sim_n` on themselves.

Using the same dose level list and toxicity assumption in the previous example, and set `sim_time` to 10.

```
data dummy_tox;
  dose=1 ;true_tox=0.06;output;
  dose=5 ;true_tox=0.13;output;
  dose=10;true_tox=0.24;output;
  dose=12;true_tox=0.30;output;
  dose=15;true_tox=0.63;output;
run;

%blrm_sim_n(dose_list=%str(1,5,10,12,15),ref_dose=10,true_tox=dummy_tox,
  start_dose=10,
  max_sample=12,cohortn=3,skip=0,
  mu1=-2.5, mu2=0, v1=4, v2=1, rho=0 ,
  alpha=0.05, nbi=500, nmc = 5000,
```

```
ud=0.16,od=0.33,ewoc=0.3,
seed=4345,sim_time=10,debug=0,time_sum=0);
```

If parameter time_sum is set to 1, then simulation time summary will be output in the ODS.

```
ods exclude close;
proc print data=WORK.TIME_SUMMARY;
  where missing(sim_batch);
  var total_start total_end total_elapsed_sec avg_sec_per_sim;
run;
title;
```

Obs	total_start	total_end	total_elapsed_sec	avg_sec_per_sim
1	29JUL2025:13:06:35	29JUL2025:13:07:06	30.9480	3.09480

Here we set the target range as [0.25,0.33), therefore only 12mg will be considered as target dose.

```
%blrm_sim_sum(
  sim_data = sim_all,
  dlt_data = dlt_all,
  true_tox = dummy_tox,
  target_range_low = 0.25,
  target_range_high = 0.33, /* Used for both target upper bound and overdose threshold */
  acceptable_low = 0.15, /* Minimum acceptable toxicity probability */
  out_summary = sim_results,debug=0
);
ods exclude close;
proc print data=sim_results label;
  format Value 6.3;
  title "Trial Simulation Summary";
run;
```

Trial Simulation Summary		
Obs	Metric	Value
1	Prop_Correct_Dose	0.333
2	Prop_Acceptable_Dose	0.667
3	Prop_Over_Toxic_Dose	0.111
4	Avg_Patients_Over_Toxic	0.900

Obs	Parameters
1	Acceptable: >=0.15 Target:0.25-0.33 Over: >0.33
2	Acceptable: >=0.15 Target:0.25-0.33 Over: >0.33
3	Acceptable: >=0.15 Target:0.25-0.33 Over: >0.33
4	Acceptable: >=0.15 Target:0.25-0.33 Over: >0.33

Appendix

Comparison to blrm package in R

In this part we compare the output between blrm package and macro blrm_main .

Assumption

- Dose list: 2,4,6,8,12
- reference dose: 12
- target toxicity interval: [0.15, 0.35)
- EWOC boundary: 0.25

DLT assumption:

Dose	Enrollment	DLT
2	6	1
4	3	0
6	0	-
8	0	-
12	0	-

Prior distribution:

$$\begin{pmatrix} \log(\alpha) \\ \log(\beta) \end{pmatrix} \sim \text{BVN} \left[\begin{pmatrix} -0.9 \\ 0 \end{pmatrix}, \begin{pmatrix} 4 & 0 \\ 0 & 1 \end{pmatrix} \right]$$

Results

%Blrm_main				blrm package		
Dose	Under dose	Target toxicity	Over dose	Under dose	Target toxicity	Over dose
2	0.7996	0.1849	0.0155	0.78782	0.19692	0.01525
4	0.596725	0.337175	0.0661	0.58773	0.34203	0.07025
6	0.45635	0.400825	0.142825	0.4572	0.39258	0.15022
8	0.377675	0.399	0.223325	0.37543	0.39368	0.2309
12	0.293475	0.35905	0.347475	0.28053	0.36755	0.35192

Other notes

In macro `blrm_main`, each time new data arrives, the macro updates the model using all the cumulative data and starts from the initial prior distribution. For user who familiar with Bayesian method will notice that, under the assumption of **exchangeability**, a simpler way should be using only the **new data** and the previous posterior distribution as the prior to update the model. The reason in `blrm_main` macro not doing so is that, unlike many Bayesian packages in R, the PROC MCMC in SAS cannot save the information of posterior distribution into an `object` and use it as a prior distribution when new data comes, instead, PROC MCMC can only use standard distribution as prior distribution.

Codes

Comparison part

- R code:

```
library(blrm)
#scenario 1
mean <- c(-0.9,0)
se <- c(2,1)
corr <- 0
prior <- list(mean = mean,
```

```

        se = se,
        corr = corr)

#parameters
seeds <- 1:99
nsamples <- 40000
burn_in <- 0.2
drug_name <- "S1"
dose_unit <- "mg"
prov_dose <- c(2,4,6,8,12)
ref_dose <- 12
category_bound <- c(0.15, 0.35)
category_name <- c("under-dosing", "targeted-toxicity", "over-dosing")
ewoc <- 0.25

# observed cohorts
dose <- c(2,4)
n_pat <- c(6,3)
dlt <- c(1,0)

# combine prior, parameters, and observed cohorts to a list
data_s1 <- list(seeds=seeds, nsamples=nsamples, burn_in=burn_in, drug_name=drug_name,
               dose_unit=dose_unit, prov_dose=prov_dose, ref_dose=ref_dose,
               dose=dose, n_pat=n_pat, dlt=dlt, category_bound=category_bound,
               category_name=category_name, ewoc=ewoc)

trial_s1 <- blrm_mono_ss(prior=prior, data=data_s1, output_excel=TRUE, output_pdf=FALSE)

```

- SAS code:

```

proc delete data=_all_;
run;
OPTIONS SOURCE SOURCE2;

data dlt_s1;
  dose=2;n=6;dlt=1;output;
  dose=4;n=3;dlt=0;output;
run;
%blrm_main(datain=dlt_s1,dataout=out_mcmc, ref_dose=12,
mu1=-0.9, mu2=0, v1=4, v2=1, rho=0 ,
alpha=0.05, nbi=2000, nmc = 40000, seed = 9527,SIMMING=0);
%blrm_ewoc(dose_list=%str(2,4,6,8,12),
           ref_dose=12,mcmcout=out_mcmc,dataout=next_dose,
           ud=0.15,od=0.35,ewoc=0.25,debug=0)

```

References

- Guo, Xiaohan, Sean Kent, Arnab Maity, and Wei Zhong. 2025. "Optimization of EWOC Principle in BLRM Design for Phase 1 Oncology Trials." *Journal of Biopharmaceutical Statistics* 35 (3): 407–23. <https://doi.org/10.1080/10543406.2024.2333530>.
- Neuenschwander, Beat, Michael Branson, and Thomas Gsponer. 2008. "Critical Aspects of the Bayesian Approach to Phase I Cancer Trials." *Statistics in Medicine* 27 (13): 2420–39. <https://doi.org/10.1002/sim.3230>.
- Neuenschwander, Beat, Satrajit Roychoudhury, and Heinz Schmidli. 2016. "On the Use of Co-Data in Clinical Trials." *Statistics in Biopharmaceutical Research* 8 (3): 345–54. <https://doi.org/10.1080/19466315.2016.11>

74149.

- Widmer, Lukas A., Andrew Bean, David Ohlssen, and Sebastian Weber. 2023. “Principled Drug-Drug Interaction Terms for Bayesian Logistic Regression Models of Drug Safety in Oncology Phase I Combination Trials.” arXiv. <https://doi.org/10.48550/arXiv.2302.11437>.
- Zhang, Hongtao, Alan Y. Chiang, and Jixian Wang. 2022. “Improving the Performance of Bayesian Logistic Regression Model with Overdose Control in Oncology Dose-Finding Studies.” *Statistics in Medicine* 41 (27): 5463–83. <https://doi.org/10.1002/sim.9402>.
- Zhao, Wei, and Harry Yang. n.d. “Statistical Methods in Drug Combination Studies.”
- Zhou, Heng, Ying Yuan, and Lei Nie. 2018. “Accuracy, Safety, and Reliability of Novel Phase I Trial Designs.” *Clinical Cancer Research* 24 (18): 4357–64. <https://doi.org/10.1158/1078-0432.CCR-18-0168>.